AN OMEGA-CATEGORICAL SUPERSIMPLE THEORY WHOSE FIBRE DIMENSION STRICTLY EXCEEDS THE NON-MINIMALITY DEGREE

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Allowed standard prerequisites (editorial).

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JB was partially supported by Simons travel grant G3535. JF was partially supported by NSF CAREER award number 1945251. SM, who is additionally affiliated with the Institut de mathématiques de Jussieu – Paris Rive Gauche, was supported by the NSF under Grant No. DMS-2303034, and by the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie grant agreement No 101034255.

- Evans's construction of *-generic structures from a class C_f with predimension, and the criteria for the generic to be well-defined and simple. Used inside the counted proof at lines 1758-1766; the individual amalgamation check follows Remark 3.8 of the cited Evans paper. Produces the theory S_n and justifies that the generic exists and is simple even though the criteria hold only for x >= n+2.
- The relation between SU-rank, non-minimality degree and predimension in this setting. Used inside the counted proof at lines 1769-1771 and 1778-1781. Turns the rank computation and the predimension estimates into the two bounds nmdeg(p) <= n and nmdeg(p) >= n.

Theorem 0.1. Let $n < \omega$. There is an ω-categorical supersimple theory S_n admitting a type p with $\text{nmdeg}(p) = n$ and $F_{ind}(p) > \text{nmdeg}(p)$.

Proof. Let $\mathcal{L} = \{R\}$ for R an k-ary relation with $k = n+1$, and $\delta(A) = 2|A| - \frac{|R(A)|}{k!}$. Then let S_n be the theory of the *-generic M for the class $\mathcal{C}_f$ where $f(x) = 2x - 2$ for $x \leq n + 2$, and $f(x) = 2(n + 2) - 2 + (\frac{\log(x)}{\log(3)} - \frac{\log(n+2)}{\log(3)})$ for $x \geq n + 2$.

Now the criteria for simplicity and well-definedness of the $*$-generic of $\mathcal{C}_f$, $0 < f'(x) < \frac{1}{x}$ and $f(3x) \leq f(x) + 1$, will hold for all $x \geq n + 2$, but not for all $x \geq 0$. However the $*$-generic of $\mathcal{C}_f$ is still well-defined and simple. As in Remark 3.8 of [1], we may individually check the instances of amalgamation where some of the structures being amalgamated have size at most $n + 1$, since these structures either have no relations, or have size $n + 1$ and one relation. Moreover, because $f(x) \geq 2$ for $x \geq k$, we again see that there is only one type $p \in S^1(\emptyset)$. And because $f(x) \leq 2x - 1$ for $x \leq k$, so there is again a structure in $\mathcal{C}_f$ of size k with exactly one instance of R up to permutation, $\text{SU}(p) = 2$ with $\text{nmdeg}(p) \leq k - 1$, so here $\text{nmdeg}(p) \leq n + 1 - 1 = n$.

To see $\text{nmdeg}(p) \geq n$, so $\text{nmdeg}(p) = n$, we show there is a unique type in $S^n(\emptyset)$ realized by distinct $a_1, \ldots, a_n$. Because $f(x) > 2n \geq d_0(a_1, \ldots, a_n)$ for $x > n + 1$ $|\text{acl}(a_1, \ldots, a_n)| \leq n+1$. But $|\text{acl}(a_1 \ldots a_n)| = n+1$ is impossible, because the least predimension of a set of size $n+1$ is $2n+1$, which is larger than $d_0(a_1 \ldots a_n) = 2n$.

Now to show that $n = \text{nmdeg}(p) < F_{ind}(p)$, note that if $\{a_i\}_{i<\omega}$ is a nonconstant indiscernible sequence, either every $n + 1$-tuple satisfies R or none does. The first is clearly impossible since long enough such sequences have negative predimension. In the second case, $d_0(a_1 \ldots a_{n+1}) = 2n + 2$. Because $f(x) > 2n + 2$ for $x > n + 2$, $|\text{acl}(a_1 \ldots a_{n+1})| \leq n+2$, so either $a_1, \ldots a_{n+1}$ is closed or $|\text{acl}(a_1 \ldots a_{n+1})| = n+2$ and $d_0(\text{acl}(a_1 \ldots a_{n+1})) \geq f(n + 2) = 2n + 2$. For either option $a_{n+1} \bigcup a_1, \ldots a_n$ and $F_{ind}(p) > n$. □

References

[1] David Evans. $\aleph_0$ -categorical structures with a predimension. Annals of Pure and Applied Logic - APAL, 116:157–186, 08 2002.